# COGS 9 Section

TA: Lindsey Gu

Friday, October 2

# Who am I

**Lindsey Gu**

2nd year Master’s student in Computer Science

- B.S. Cogsci Design & B.A. ICAM (visual art) @ UC San Diego

Hobbies: interactive/immersive media, photography, musicals, playing bamboo flute, and anything weird ;)

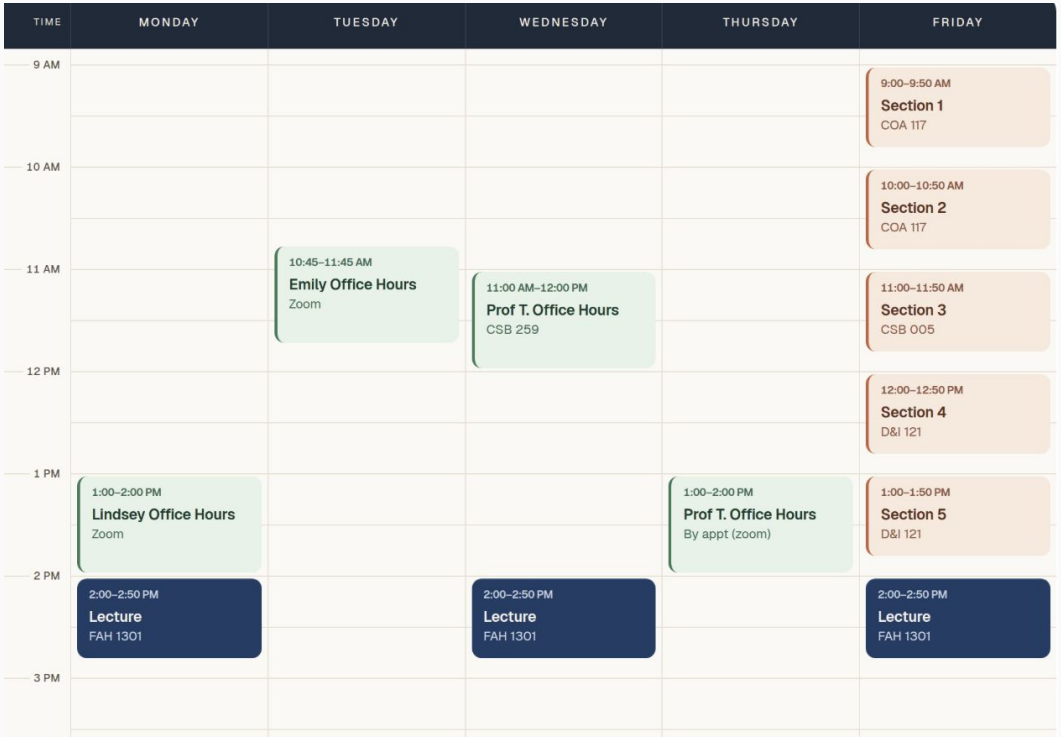
My personal website: www.yuemenglindseygu.com



# Office Hours

TA office hours are **Mondays/Tuesdays**

- Posting questions: **Piazza**
- Please send an email for further questions, no Canvas messages!



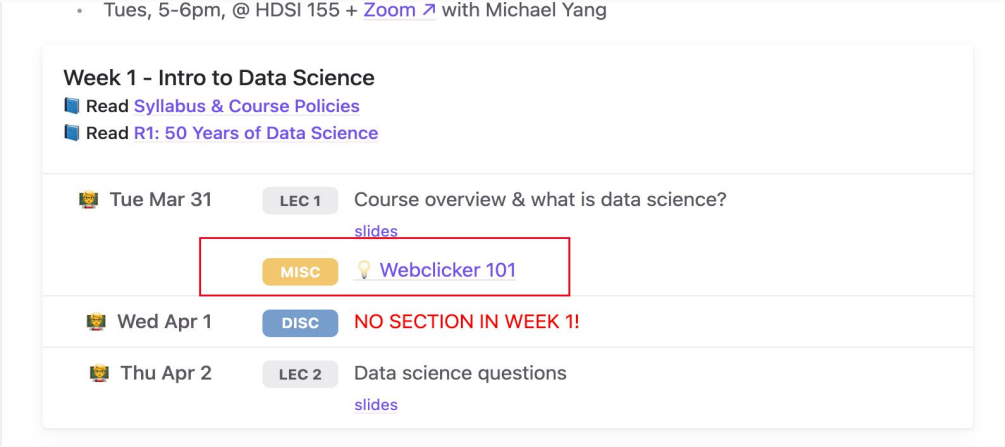
# Reminders

- **Due *Wednesday* @ 11:59** *(can all be found on course website)*
  - Precourse Survey
  - Reading quiz 1
  - Syllabus quiz
  - If you miss assignments, you can still submit with late penalty (better than nothing!)

- If you are joining late, send Prof. Tomoschuk + us an email and you have until Friday of Week 3 to submit missed assignments

# Webclicker

We use it for attendance.



- Tues, 5-6pm, @ HDSI 155 + Zoom ↗ with Michael Yang

**Week 1 - Intro to Data Science**

📘 Read Syllabus & Course Policies

📘 Read R1: 50 Years of Data Science

| | | |
|---|---|---|
| Tue Mar 31 | LEC 1 | Course overview & what is data science? slides |
| | MISC | 💡 Webclicker 101 |
| Wed Apr 1 | DISC | NO SECTION IN WEEK 1! |
| Thu Apr 2 | LEC 2 | Data science questions slides |

# Attendance Webclicker

webclicker.web.app




Code: OMFZUC

Different from lecture code!!

If you have issues using webclicker, please let us know before you leave the section!

Are you ready for Week 2?

A. 😀 B. 😎 C. 🤔 D. 😅 E. 😴

# Questions?

# The goals

Another space to meet and connect with your classmates

For **group projects**, you will be in groups of 3-5 students from the same section you can regularly attend

Discuss readings and related topics

Share ideas and learn from each other

# The project

“Your task is to propose a project that tackles a novel question that can be answered with data, and to describe how you would integrate multiple data science methods and ideas to address it.”

Group size: 4-5
Formed by **END OF WEEK 3** - you can choose your own, or we can match you (form next week)

# Question Example

- Do basketball players with more rest days score higher points per game than players with fewer rest days?

- Topic:
- Outcome or Target:
- Features:
- Compared to what?:

# Question Example

- Do **basketball players** with **more rest days** score **higher points per game** than **players with fewer rest days**?

- **Topic:** Basketball Players
- **Outcome or Target:** Points per game
- **Features:** More rest days
- **Compared to what?:** Players with fewer rest days

# Improved Question Example

- Do **basketball players** with **more rest days** score **higher points per game** than **players with fewer rest days**?
- Do **NBA players** during **the 2025–26 regular season** with at least **2 rest days** before a game score **higher points per game** than **players on back-to-backs**?

- **Topic, Timeframe, Features, Outcome, Comparison Group**

# Meet & greet

Now, it’s time to get to know your classmates!

Talk to your neighbours (groups of 3-4) about yourself (name, year, major, etc.)

As a group, find three UNIQUE things you have in common. For example:

- Something you did over the summer ;)
- A goal you want to achieve this quarter

We will invite some groups to share afterward!

# Common Thread

Some ideas to help out…

- Favorite food 🍔
- A show, movie, or YouTube channel you all like 🎬
- A place you've visited or want to visit ✈️
- Music genre or artist you all like 🎵
- A hobby you enjoy 🎨
- Preferred study spot (library, home, café, etc.) 📝
- A pet you have or want 🐶
- A topic you want to do for the project later

# TIME FOR A GAME!

# This or That!

Data science is just a rebranding of statistics." (Discuss with your group – guilty, not guilty, or it's complicated?)

- Use the reading in your response

Code: OMFZUC

webclicker.web.app

**WEBCLICKER:**

**A. True**
**B. False**
**C. Complicated**

# This or That!

Data science is just a rebranding of statistics." (Discuss with your group — guilty, not guilty, or it's complicated?)

- Use the reading in your response

Code: OMFZUC

webclicker.web.app

**WEBCLICKER:**

A. **True**
B. **False**
C. **Complicated**

**Complicated.** Data science grew out of statistics, but Donoho argues that it **expands beyond traditional statistics** by including computing, data wrangling, prediction, visualization, reproducibility, and science about data science

# This or That!

Big data is what fundamentally separates data science from statistics.

- Use the reading in your response

**WEBCLICKER:**

**A. True**
**B. False**
**C. Complicated**

Code: OMFZUC

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# This or That!

Big data is what fundamentally separates data science from statistics.

- Use the reading in your response

**WEBCLICKER:**

**A. True**
**B. False**
**C. Complicated**

**False,** big data alone does not distinguish data science from statistics. What makes data science broader is **its emphasis on the full workflow of learning from data**

Code: OMFZUC

webclicker.web.app

# This or That!

Tukey believed data analysis was a branch of mathematics

- Use the reading in your response

**WEBCLICKER:**

**A. True**
**B. False**
**C. Complicated**

Code: OMFZUC

webclicker.web.app

# This or That!

Tukey believed data analysis was a branch of mathematics

- Use the reading in your response

**WEBCLICKER:**

**A. True**
**B. False**
**C. Complicated**

**False.** Tukey argued that data analysis was broader than mathematical statistics and should be treated as its own science, centered on practical methods for learning from data.

Code: OMFZUC

webclicker.web.app

# R1 discussion

## 50 Years of Data Science

David Donoho

Department of Statistics, Stanford University, Standford, CA

**ABSTRACT**

More than 50 years ago, John Tukey called for a reformation of academic statistics. In "The Future of Data Analysis," he pointed to the existence of an as-yet unrecognized *science*, whose subject of interest was learning from data, or "data analysis." Ten to 20 years ago, John Chambers, Jeff Wu, Bill Cleveland, and Leo Breiman independently once again urged academic statistics to expand its boundaries beyond the classical domain of theoretical statistics; Chambers called for more emphasis on data preparation and presentation rather than statistical modeling; and Breiman called for emphasis on prediction rather than inference. Cleveland and Wu even suggested the catchy name "data science" for this envisioned field. A recent and growing phenomenon has been the emergence of "data science" programs at major universities, including UC Berkeley, NYU, MIT, and most prominently, the University of Michigan, which in September 2015 announced a $100M "Data Science Initiative" that aims to hire 35 new faculty. Teaching in these new programs has significant overlap in curricular subject matter with traditional statistics courses; yet many academic statisticians perceive the new programs as "cultural appropriation." This article reviews some ingredients of the current "data science moment," including recent commentary about data science in the popular media, and about how/whether data science is really different from statistics. The now-contemplated field of data science amounts to a superset of the fields of statistics and machine learning, which adds some technology for "scaling up" to "big data." This chosen superset is motivated by commercial rather than intellectual developments. Choosing in this way is likely to miss out on the really important intellectual event of the next 50 years. Because all of science itself will soon become data that can be mined, the imminent revolution in data science is not about mere "scaling up," but instead the emergence of scientific studies of data analysis science-wide. In the future, we will be able to predict how a proposal to change data analysis workflows would impact the validity of data analysis across all of science, even predicting the impacts field-by-field. Drawing on work by Tukey, Cleveland, Chambers, and Breiman, I present a vision of data science based on the activities of people who are "learning from data," and I describe an academic field dedicated to improving that activity in an evidence-based manner. This new field is a better academic enlargement of statistics and machine learning than today's data science initiatives, while being able to accommodate the same short-term goals. *Based on a presentation at the Tukey Centennial Workshop, Princeton, NJ, September 18, 2015.*

**ARTICLE HISTORY**
Received August 2017
Revised August 2017

**KEYWORDS**
Cross-study analysis; Data analysis; Data science; Meta analysis; Predictive modeling; Quantitative programming environments; Statistics

## R1: Reading Guide

*Donoho, D "50 Years of Data Science"*

### Why we're reading this first

This paper is a "big-picture map" of data science: where it came from, what problems it was trying to solve, and why it isn't just a rebranding of statistics or machine learning. Donoho's main goal is to give you a framework for thinking about data science as a discipline, not just a set of tools.

Key theme: data science emerged partly as a response to gaps in traditional statistics education and practice, especially around computation, data wrangling, and real-world workflows. Donoho argues that calling data science merely "statistics + machine learning" misses what makes it distinct.

### How to read this paper (so it doesn't feel like a wall of text)

A useful strategy

Read once for structure, then reread the key framework sections:

- First pass: identify the *big claims* and *the framework*.
- Second pass: focus on the six divisions and the examples he uses to justify them.

What to look for as you read

1. What problem is Donoho trying to solve? (Hint: confusion about what data science is and what it should include.)
2. What counts as "data science work" that older framings ignored or undervalued?
3. Where does he draw boundaries between statistics, machine learning, and data science, and why?
4. What does it mean to make data science a "science" rather than just a set of skills?

By the end, you should be able to explain:

- why Donoho thinks data science needed a framework beyond statistics
- the six divisions and a real example of each
- what CTF is, and one benefit + one limitation
- what "science about data science" is and why it matters for trust in results

# R1 discussion

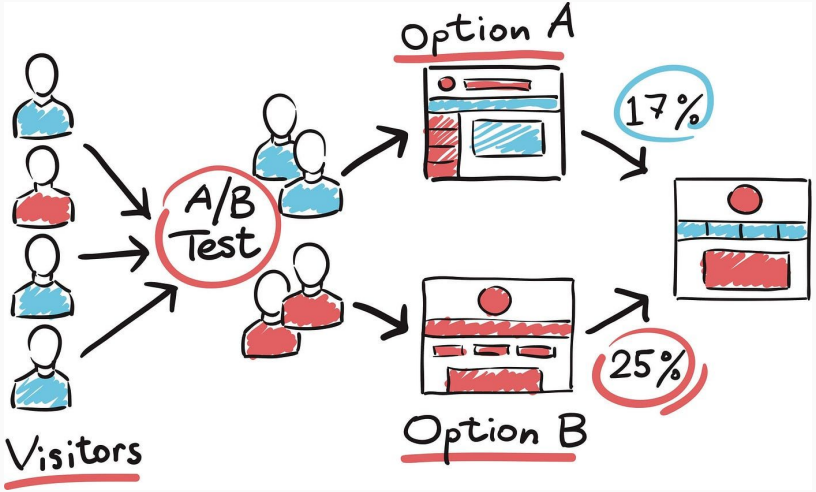
"Greater Data Science" in six divisions

- **Data exploration and preparation:** Finding patterns, anomalies, missingness, and forming early hypotheses.
- **Data representation and transformation:** Cleaning, reshaping, feature construction, merging sources—turning raw data into analyzable data.
- **Computing with data:** Programming, pipelines, scalability, reproducibility, and tools/infrastructure.
- **Data modeling:** Statistical modeling and machine learning (prediction, inference, uncertainty, evaluation).
- **Data visualization and presentation**: Communicating results and building understanding through visuals and narrative.
- **Science about data science:** Studying how data analysis itself works: workflow variation, reproducibility, benchmarking, and reliability.

# Data Driven Decisions: A/B Testing

**A/B testing**

A/B testing is employed to compare multiple versions of a single variable, for example by testing a subject's response to variant A against variant B, and to determine **which of the variants is more effective.**



# Examples of Different A/B Tests

## Messaging

Buy Now

Check Out

## Shape

Buy Now

Buy Now

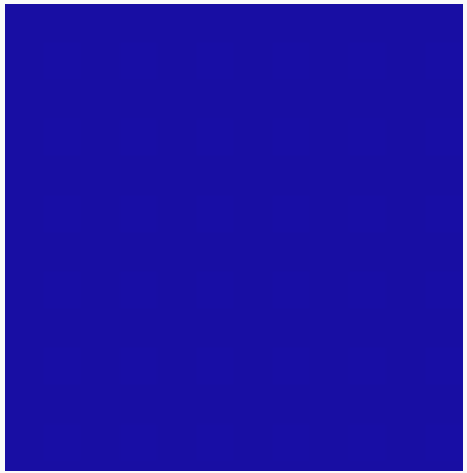
## Color

Buy Now

Buy Now

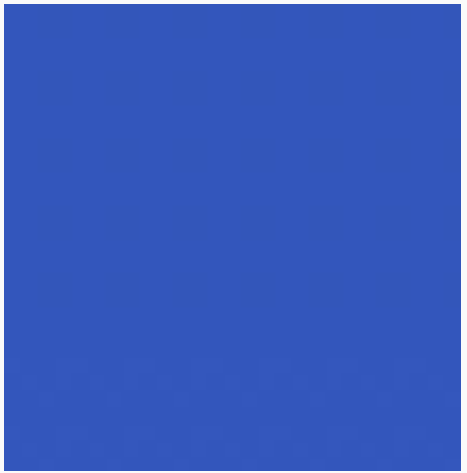
# Class Activity

**Which link color is more likely to be clicked?**



**What is Data Science?**

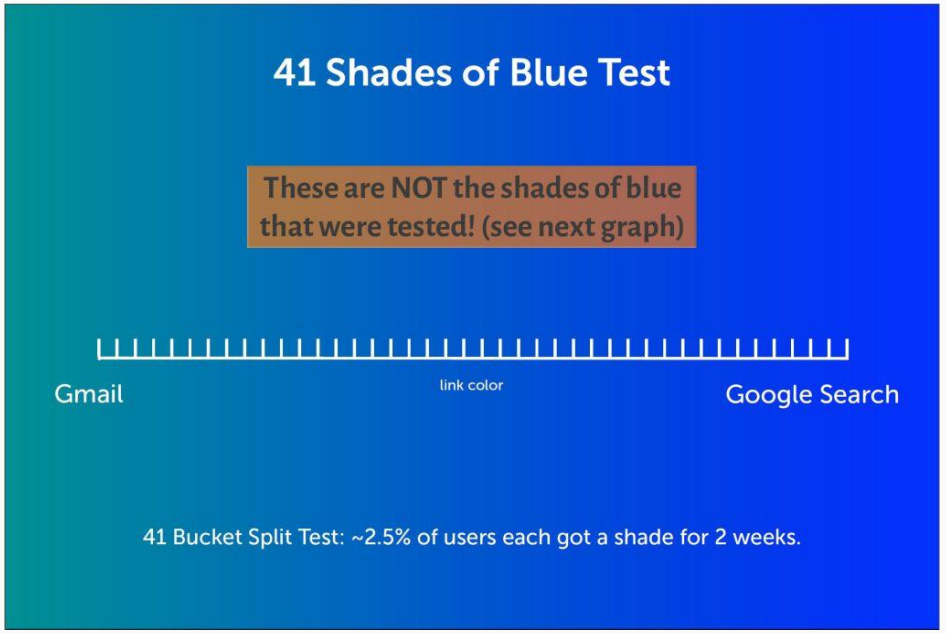
**vs.**



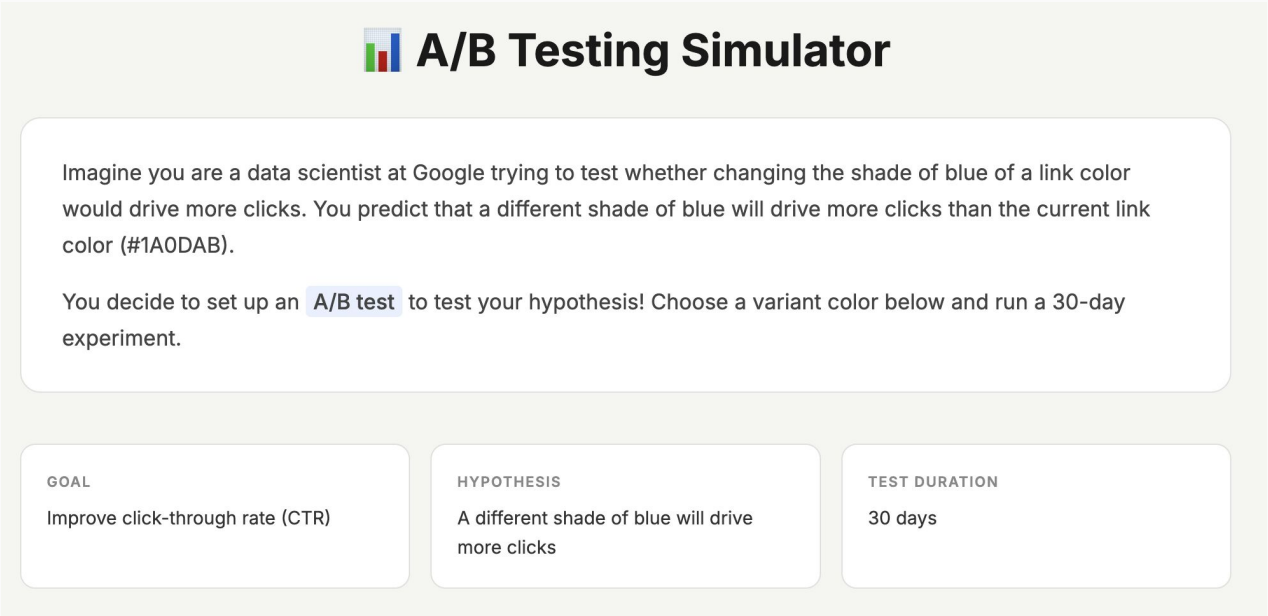
**What is Data Science?**

# Example: Google

**41 Shades of Blue Test**
~2009 Google utilized A/B testing to test 41 shades of blue and found that a slightly more purple-blue shade significantly increased clicks, generating an **extra $200 million** in annual revenue



# Demo

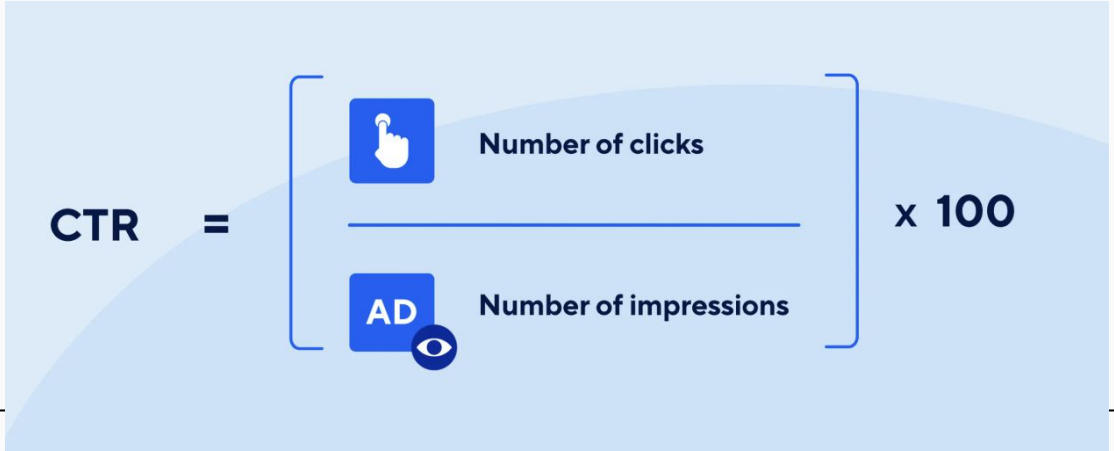



Try it out yourself here: https://bit.ly/ab-testing-demo

# Steps to A/B Testing

**Define the goal (What metric are you trying to improve?)**

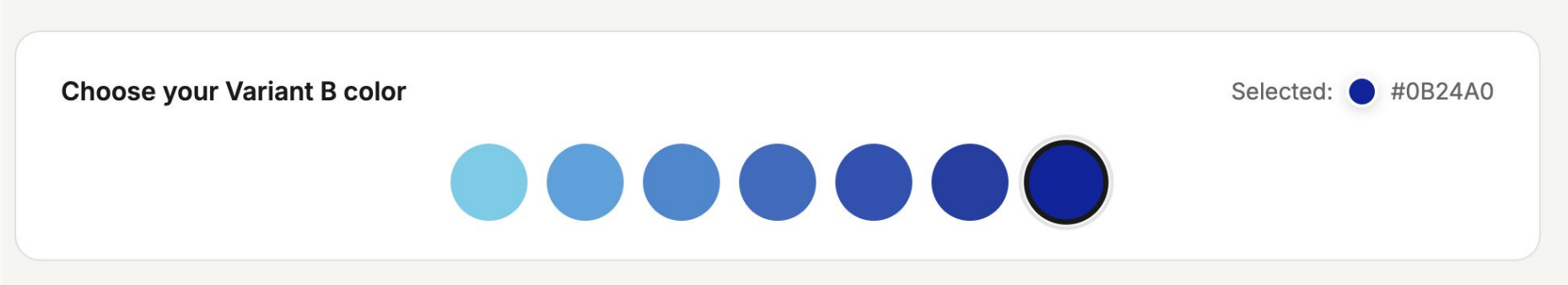
For this test we're trying to improve click-through rate (the percent of users who view the link who also click it)



# Steps to A/B Testing

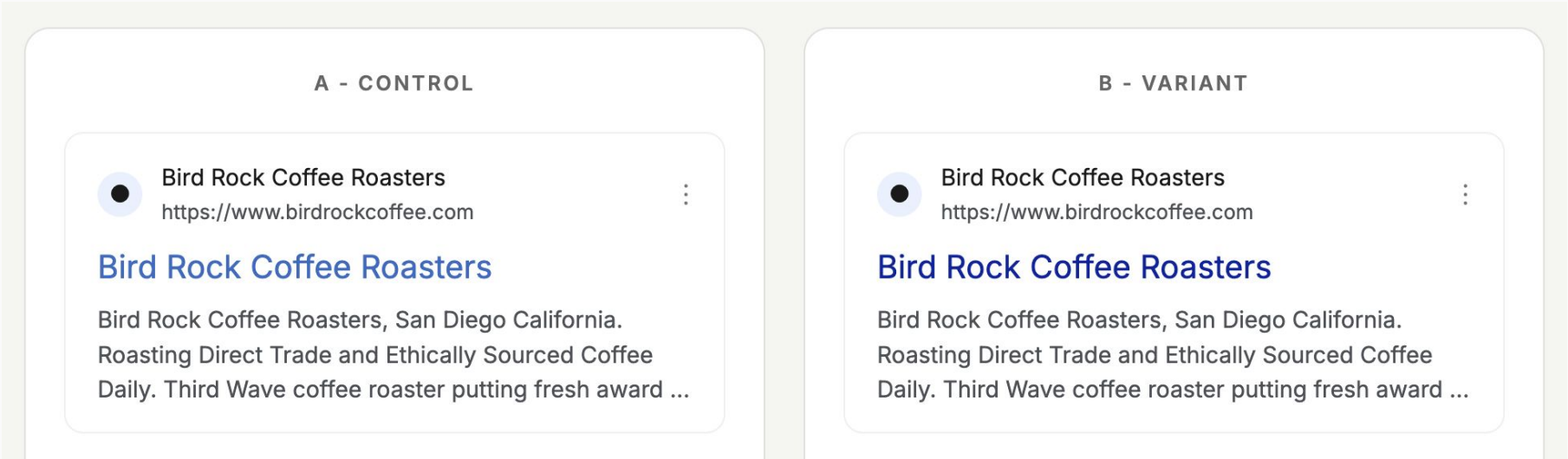
**Form a hypothesis (What change do you think will improve the click-through rate?)**

E.g: Changing the link color to a darker and more vibrant shade of blue will be more visually engaging and result in more clicks.



# Steps to A/B Testing

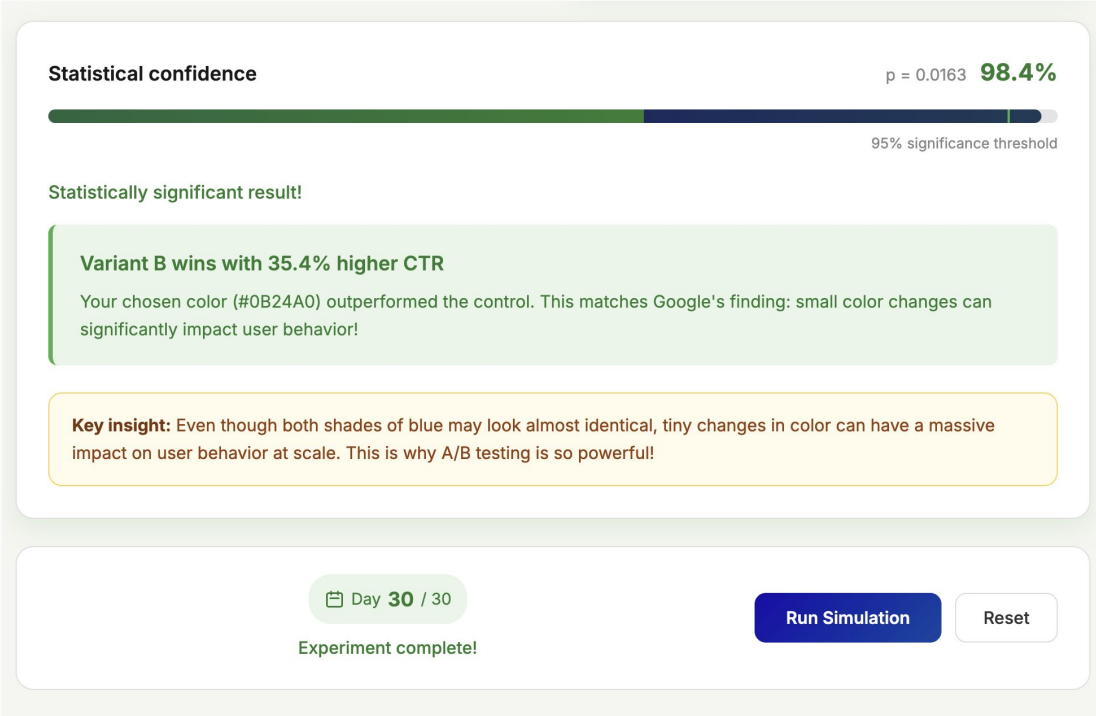
## Design the experiment (Set your control (A) vs. variant (B)



# Steps to A/B Testing

## Run and analyze (Did your variant make a difference?)

In this case, yes! Like the original case study, a darker shade of blue-purple drives more clicks.



# Questions?